30/10/2k26

basic python

In [2]:
```
age, income, saving =21,4000,2000
print("age:",age)
print("income:",income)
print("saving:",saving)
```

```
age: 21
income: 4000
saving: 2000
```

In [4]:
```
price=int(input("Enter the Input:"))
quantity=int(input("Enter the quantity:"))
total_cost=price*quantity
print(total_cost)
```

```
24
```

In [60]:
```
a=12
b=2.3
c=12+12j
d="text"
e=None
f=True
g=True+True
print(type(a))
print(type(b))
print(type(c))
print(type(d))
print(type(e))
print(type(f))
print(type(g)) #special case
print(g)
```

```
<class 'int'>
<class 'float'>
<class 'complex'>
<class 'str'>
<class 'NoneType'>
<class 'bool'>
<class 'int'>
2
```

implicit and explicit conversion

In [7]:
```
ic=200
bc=78.2
tot=ic+bc
print(type(ic))
print(type(bc))
print(type(tot))
```

```
<class 'int'>
<class 'float'>
<class 'float'>
```

In [9]:
```
price1=100
tax='18'
total_amount=price1+int(tax)
print(type(price1))
print(type(tax))
print(type(total_amount))
print(total_amount)
```

```
<class 'int'>
<class 'str'>
<class 'int'>
118
```

python operator

1. arithmetic opperation

+,-,%,*,/,//

In [19]:
```
x='100'
print(int(x))
print(float(x))
print(str(x),sep='')
print(bool(x))
```

```
100
100.0
100
True
```

In [23]:
```
x='100.02'
print(float(x))
print(str(x),sep='')
print(bool(x))
#print(int(x))
```

```
100.02
100.02
True
```

In [30]:
```
price2=100
tax2=12.0
total_cost2=price2+tax2
print('total amount:',total_cost2)
```

```
total amount: 112.0
```

In [31]:
```
price3=100
tax3=12.0
discount=12
total_cost3=price3+tax3-discount
print('total amount:',total_cost3)
```

```
total amount: 100.0
```

In [33]:
```
price4=100
tax4=12.0
quantity4=12
discount4=12
total_cost4=tax4-discount4+price4*quantity4
print('total amount:',total_cost4)
```

```
total amount: 1200.0
```

In [28]:
```
# special case in string with addition
text="Age is :"
age2='22'
combine=text+age2 #concatenate
print(combine)
```

```
Age is :22
```

In [35]:
```
#speical case in string with multiplication
text5="hi "
times=4
combine=text5*times
print(combine)
```

```
hi hi hi hi
```

In [40]:
```
#get quote(int divison)
total_cost5=100
quantity5=10
price_per_unit=total_cost5/quantity5
print(price_per_unit)
```

```
10.0
```

In [39]:
```
#get quote(floor divison)
total_cost5=1234567890
quantity5=1234
price_per_unit=total_cost5//quantity5
print(price_per_unit)
```

```
1000460
```

In [47]:
```
#to get the remainder
total_cost6=100
```

```
quantity6=10
price_per_unit=total_cost6/quantity6
remaiding_amount=total_cost6%quantity6
print(price_per_unit)
print(remaiding_amount)
```

```
10.0
0
```

2.Relation operator

<,<=,>,>=,!=,==

In [54]:
```
value1=100
value2=2
```

In [57]:
```
print('1.',value1<value2)
print('2.',value1<=value2)
print('3.',value1>value2)
print('4.',value1>=value2)
print('5.',value1==value2)
print('6.',value1!=value2)
```

```
1. False
2. False
3. True
4. True
5. False
6. True
```

3.logical operator

AND,OR,NOT

In [61]:
```
x1=True
y1=False
```

In [62]:
```
print(x1 and y1)
```

```
False
```

In [63]:
```
print(x1 or y1)
```

```
True
```

In [67]:
```
print(not x1)
```

```
False
```

4. membership operator

in ,not in

In [76]:
```
string='mahesh'
var='a'
print(var in string)
#print(string in var)
```

```
True
```

In [75]:
```
print(string not in var)
```

```
True
```

In [78]:
```
string1='mahesh'
var1='z'
print(var1 not in string1)
```

```
True
```

5.bitwise operator

& , | , ~ , >> , << , ^

In [5]:
```
a2=14
b2=13
```

In [6]:
```
print(a2 & b2)
```

```
12
```

In [7]:
```
print(a2 | b2)
```

```
15
```

In [9]:
```
print(a2 ^ b2)
```

```
3
```

In [12]:
```
print(~a2)
```

```
-15
```

In [13]:
```
print(a2>>1)
```

```
7
```

In [14]:
```
print(a2<<1)
```

```
28
```

In [ ]: